Pregătire Examen MN 2017

2013 – 14 iunie – B

1. Spline C_2

Natural

$$A = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ h_0 & 2(h_0+h_1) & h_1 & & \vdots \\ 0 & h_1 & 2(h_1+h_2) & h_2 & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ & & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ 0 & \cdots & \cdots & 0 & 1 \end{bmatrix}$$

$$x = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ \frac{3}{h_1}(a_2-a_1) - \frac{3}{h_0}(a_1-a_0) \\ \vdots \\ \frac{3}{h_{n-1}}(a_n-a_{n-1}) - \frac{3}{h_{n-2}}(a_{n-1}-a_{n-2}) \\ 0 \end{bmatrix}$$

Tensionat

$$A = \begin{bmatrix} 2h_0 & h_0 & 0 & \cdots & 0 \\ h_0 & 2(h_0+h_1) & h_1 & & \vdots \\ 0 & h_1 & 2(h_1+h_2) & h_2 & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ & & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ 0 & \cdots & \cdots & h_{n-1} & 2h_{n-1} \end{bmatrix}$$

$$x = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} \quad b = \begin{bmatrix} \frac{3}{h_0}(a_1-a_0) - 3f'(a) \\ \frac{3}{h_1}(a_2-a_1) - \frac{3}{h_0}(a_1-a_0) \\ \vdots \\ \frac{3}{h_{n-1}}(a_n-a_{n-1}) - \frac{3}{h_{n-2}}(a_{n-1}-a_{n-2}) \\ 3f'(b) - \frac{3}{h_{n-1}}(a_n-a_{n-1}) \end{bmatrix}$$

Condiții

Natural	Tensionat
$S''_0(x_0) = 0$	$S'_0(x_0) = f'(x_0)$
$S''_{n-1}(x_n) = 0$	$S'_{n-1}(x_n) = f'(x_n)$

2. Nu se face

3. Spline tensionat Q^2

$f(x): x = [0, 1, 2]$, $f = [6, 4, 3]$,

$f' = [1 \ * \ -1]$

$$\begin{cases} S_0(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 \\ S_0'(x) = b_0 + 2c_0(x - x_0) + 3d_0(x - x_0)^2 \end{cases}$$

$$\begin{cases} S_1(x) = a_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3 \\ S_1'(x) = b_1 + 2c_1(x - x_1) + 3d_1(x - x_1)^2 \end{cases}$$

$S_0(x_0) = f(x_0)$

$S_0(0) = 6$

$\boxed{a_0 = 6}$

$S_1(x_1) = f(x_1)$

$S_1(1) = 4$

$\boxed{a_1 = 4}$

$S_0(x_1) = S_1(x_1)$

$6 + b_0(1) + c_0(1) + d_0(1) = 4$

$\boxed{b_0 + c_0 + d_0 = -2}$

$S_0'(x_1) = S_1'(x_1)$

$b_0 + 2c_0(1) + 3d_0(1) = b_1$

$\boxed{b_0 + 2c_0 + 3d_0 - b_1 = 0}$

$S_0''(x_1) = S_1''(x_1)$

$2c_0 + 6d_0(x_1 - x_0) = 2c_1$ ($x_1 - x_0 = 1$)

$\boxed{2c_0 + 6d_0 - 2c_1 = 0}$

$a_0\ b_0\ c_0\ d_0 \mid a_1\ b_1\ c_1\ d_1$

6	1				
4					

$S_0'(x_0) = f'(x_0)$

$\boxed{b_0 = 1}$

$S_1'(x_2) = f'(x_2)$

$b_1 + 2c_1(1) + 3d_1(1) = 1$

$\boxed{b_1 + 2c_1 + 3d_1 = 1}$

$S_1(x_2) = f(x_2)$

$4 + b_1(1) + c_1(1) + d_1(1) = 3$

$\boxed{b_1 + c_1 + d_1 = -1}$

$b_1 + 2c_1 + 3d_1 = 1$ $\quad / \cdot 0$

$\boxed{c_1 + 2d_1 = 2}$

$\boxed{c_0 + d_0 = -2 - b_0 = -3}$

$\boxed{2c_0 + 3d_0 = b_1 - b_0 = b_1 - 1}$

$3d_0 = 2c_1 - b_1 + 1$

$2c_0 + 6d_0 = 2c_1$

a_0	b_0	c_0	d_0	a_1	b_1	c_1	d_1			
1	0	0	0	0	0	0	0	a_0	6
0	1	0	0	1	0	0	0	b_0	1
0	1	1	1	0	0	0	0	c_0	-2
0	1	2	3	0	-1	0	0	d_0	$=$ 0
0	0	2	6	0	0	-2	0	a_1	0
0	1	0	0	0	0	0	0	b_1	1
0	0	0	0	0	1	2	3	c_1	1
0	0	0	0	0	1	1	1	d_1	-1

$a_0 = 6$

$a_1 = 4$

$b_0 = 1$

$b_0 + c_0 + d_0 = -2 \equiv c_0 + d_0 = -3$

$b_0 + 2c_0 + 3d_0 - b_1 = 0 \equiv 2c_0 + 3d_0 - b_1 = -1$

$2c_0 + 6d_0 - 2c_1 = 0$

$b_1 + 2c_1 + 3d_1 = 1$

$b_1 + c_1 + d_1 = -1$

~~$2c_0 + 6d_0 = 2(-1 - b_1 - d_1)$~~

$-2b_1 - c_1 = 4$

~~$= 2b$~~ $c_1 = -2b_1 - 4$

$2c_0 + 6d_0 + 2b_1 + 8 = 0$

$2c_0 + 3d_0 - b_1 = -1$

$c_0 + d_0 = -3$

$-6 + d_0 - b_1 = -1$

$-6 + 4d_0 + 2b_1 = -8$

$d_0 - b_1 = 5$

$4d_0 + 2b_1 = -2$

~~$+2d_0 + b_1 = +1$~~

~~$6d_0 = 12 \Rightarrow$~~ $\boxed{d_0 = 2}$

$2d_0 - 2b_1 = 10$

$4d_0 + 2b_1 = -2$

$6d_0 = 8$

$d_0 = 4/3$

$-6 + d_0 - b_1 = -1$

$\boxed{d_0 - b_1 = 5}$

$-6 + 4d_0 - 2c_1 = 0$

$\boxed{4d_0 - 2c_1 = 6}$

$\boxed{c_1 + 2d_1 = 2}$

$4d_0 - 4b_1 = 20$

$+ 2c_1 - 4b_1 = 14$

$c_1 = 2 - 2d_1$

$4 - 4d_1 - 4b_1 = 14$

$-4(d_1 + b_1) = 10$

$-4d_1 - 4b_1 = 10$

$4b_1 + 4d_1 = -10$

$4b_1 + 4d_1 + 4c_1 = -4$

$4c_1 = 6$

$\boxed{q = 6/4 = 3/2}$

4.7

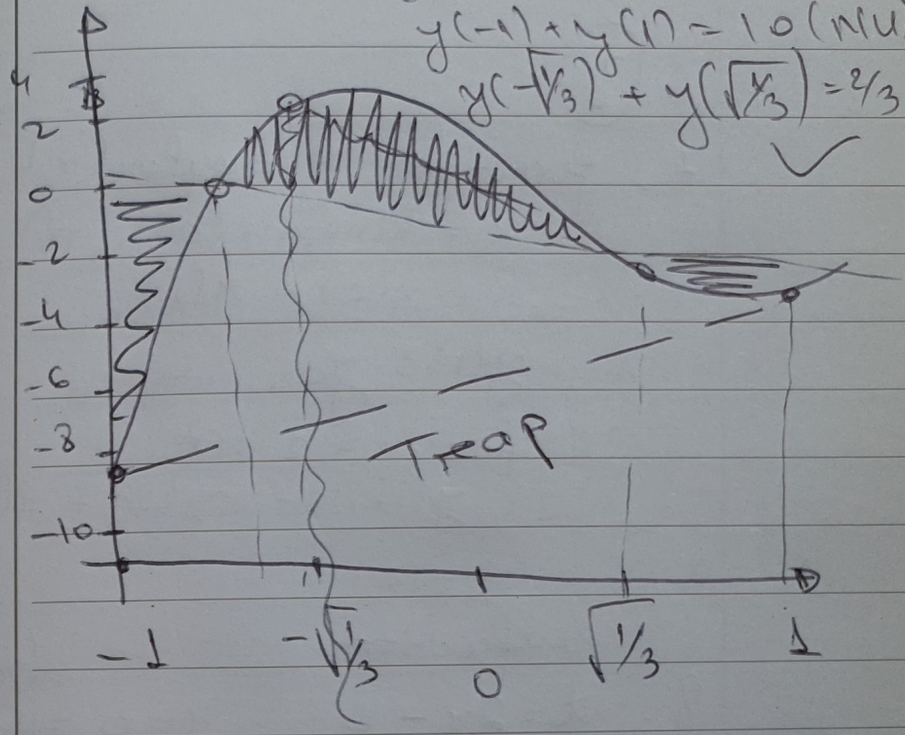
5. ~~$\langle u_1, u_1 \rangle c_1 + \langle u_2, u_1 \rangle c_1 + \dots + \langle u_n, u_1 \rangle$~~

$$\begin{cases} \langle u_1, u_1 \rangle c_1 + \langle u_2, u_1 \rangle c_2 + \dots + \langle u_n, u_1 \rangle c_n = \langle f, u_1 \rangle \\ \langle u_1, u_2 \rangle c_1 + \langle u_2, u_2 \rangle c_2 + \dots + \langle u_n, u_2 \rangle c_n = \langle f, u_2 \rangle \\ \vdots \\ \langle u_1, u_n \rangle c_1 + \langle u_2, u_n \rangle c_2 + \dots + \langle u_n, u_n \rangle c_n = \langle f, u_n \rangle \end{cases}$$

Sistem simetric și rău condiționat al cărui det. poartă numele de det. GRAM.
Rezolvare greoaie $\Rightarrow$ diferite lacune
Pt. o bază ortonormată $\Rightarrow$ sistem diagonal
, iar $c_k^* = \langle f, u_k \rangle \equiv$ coef. Fourier

4. NC folosesc nucle echidistante
Integ. gaussiană folosește nucle cu rădăcinile ale unor pol. ortogonale

$y(x) = 7x^3 - 8x^2 - 3x + 3$
$y(-1) + y(1) = 10 \; (N \setminus U)$
$y(-\sqrt{1/3}) + y(\sqrt{1/3}) = 2/3$



O cuadratură gaussiană de n nucle aproximează exact polinoame de grad $2n-1$ sau mai mic

6. ?

7. function y = euler(a, b, f, y0, n)
    x = linspace(a, b, n);
    h = x(2) - x(1);
    y(1) = y0;
    for i = 1 : n
        y(i+1) = y(i) + f(x(i), y(i))*h;
    endfor
endfunction